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Evaluation of the Canadian fire weather index system in an eastern Mediterranean environment

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ABSTRACT: The Fire Weather Index module of the Canadian Forest Fire Danger Rating System (CFFDRS) was evaluated during two consecutive fire seasons in the Mediterranean environment of Crete, Greece. The Duff Moisture Code (DMC), the Drought Code (DC), the Buildup Index (BUI) and the Fire Weather Index (FWI) were highly correlated to fire occurrence but only moderately to area burned. Logistic regression was applied in order to classify the FWI values into fire danger classes appropriate for the Mediterranean environments, as follows: 0–38 Low, 39–48 Moderate, 49–59 High, >60 Extreme. The new classification was necessary because the existing Canadian fire danger classes were found inapt for the dry and extremely fire prone eastern Mediterranean climate of Crete. After the modification, the fluctuation of the FWI values predicted more successfully the days of high fire risk, as proved by the actual fire occurrence. High correlation was found between measured litter (L layer) moisture values and those predicted by the Fine Fuel Moisture Code (FFMC). The use of an equilibrium duff moisture content value lower than 20% in Mediterranean environments, would probably improve the Duff Moisture Code (DMC) predictions. The Drought Code (DC) was poorly correlated to the upper soil moisture content. Overall, the FWI demonstrated several aptitudes related to its potential use as a meteorological fire danger rating index in Mediterranean regions. However, long–term studies are necessary to determine the precise range of each fire danger class according to fire occurrence data. Copyright © 2010 Royal Meteorological Society

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1. Introduction

In regions with Mediterranean climate, wildland fires present a major disturbance to natural ecosystems, resulting in significant economic and ecological losses. Thus, forest fire risk assessment becomes an important component of land management because there is a great advantage to be gained from being able to anticipate in advance the probability of fire occurrence on a given date. It is also economically judicious since it requires relatively reasonable investments if compared to the costs of non-scientifically based fire suppression operations (Johnston, 1988). At the same time, significant savings can be made by adopting an existing fire danger rating system that, otherwise, need to be spent on basic research and development (Fogarty *et al*., 1998).

Fire danger is expressed as the result of both constant and variable environmental factors affecting the ignition, spread and difficulty controlling wildland fires. It is important to distinguish fire danger from fire hazard and fire risk. Fire hazard represents the potential fire behaviour of a fuel complex based on its physical and chemical properties, while fire risk is the probability of a fire starting due to the potential number of ignition sources in an area (Pyne *et al*., 1996).

Forest fire danger rating systems try to integrate meteorological data with the bio-physical properties of natural fuels in order to predict their flammability and combustibility. Fire danger rating uses weather observations at a fixed site to give a broad area assessment of fire potential. Thus, qualitative and/or quantitative numerical indices of fire potential are produced and used as criteria in a variety of fire management activities. Seasonal plots of a fire danger index value are useful in visualizing how the fire season is progressing and to compare one season to another (Pyne *et al*., 1996). Integrating the data on the number of fires that have occurred in the same area, the fire danger indices may be used to determine the duration and the peak of the fire season (Harrington *et al*., 1983).

Fire danger rating is a fire management system that integrates the facets of selected fire danger factors into one or more qualitative or numerical indices of current protection needs (Chandler *et al*., 1983). Fire danger rating systems are used by fire and land management agencies to determine levels of preparedness, to issue public warnings and to provide an appropriate scale for management, research and law for fire related matters (Cheney and Gould, 1995). All these systems integrate weather variables to assess fire danger, calculated as a numerical index. Although fire danger is not the same

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as fire behaviour, the indices were developed on the assumption that fire danger is related to fire behaviour, quantified as rate of spread or intensity. Index values are classified into rating classes to aid interpretation. Many systems use four or five rating classes: Low, Moderate, High, (Very High), and Extreme (Wotton, 2009).

The occurrence and propagation of forest fires depend on many factors, some of them natural and others related to human activity. Even in regions such as the Mediterranean Basin, where most fires are human-caused, natural conditions that affect fuel properties and especially their moisture content, play a very important role in the ignition and propagation of wildland fires (Dimitrakopoulos *et al*., 2010). Since moisture content is the most significant determinant of forest fuel ignitability and combustibility, it is reasonable to expect that this single parameter should be directly related to forest fire risk and behaviour (Tunstall, 1988). Frequent measurements of live and dead fuel moistures combined with parallel variations of meteorological parameters can be used as indicators of forest fire risk (Dimitrakopoulos and Bemmerzouk, 2003).

A variety of fire danger rating systems are used around the world, including the Canadian Forest Fire Danger Rating System (CFFDRS) used in Canada (Van Wagner, 1987), the National Fire Danger Rating System (NFDRS) used in the USA (Deeming *et al*., 1977), the McArthur Forest Fire Danger Index (FFDI) used in the eastern parts of Australia (McArthur, 1967), the Forest Fire Behaviour Tables (FFBT) developed for use in Western Australia (Sneeuwjagt and Peet, 1985), and a variety of fire danger indices used in Southern European countries.

The CFFDRS is founded from the analysis of field data of three kinds: weather, fuel moisture and test fire behaviour, all collected over several decades (Simard and Main, 1982). The FWI System (Van Wagner, 1987, Van Wagner and Pickett, 1985) was the first subsystem developed in the CFFDRS and it provides numerical ratings of relative fire potential based solely on weather observations. FWI components depend solely on daily measurements of dry-bulb temperature, air relative humidity, 10 m open wind speed and 24 h accumulated precipitation, recorded at noon (local standard time, LST) (Turner and Lawson, 1978).

The FWI System component values are valid for the heat of the day (approximately 1500–1700 LST). Computation depends on the previous day's output, so daily readings must be taken. The Fire Weather Index module (FWI System) of the CFFDRS provides numerical rating of relative wildland fire potential in a standard fuel type on level terrain. It is composed of six components that individually and collectively account for the effects of fuel moisture and wind on fire behaviour (Van Wagner, 1987).

- There are three moisture codes. (1) The Fine Fuel Moisture Code (FFMC) is a numerical rating of the moisture content of litter and other cured fuels. This code is an indicator of the relative ease of ignition and flammability of fine fuel in a forest stand in a layer of dry weight about 0.25 kg m^{-2}. (2) The Duff Moisture Code (DMC) is a numerical rating of the average moisture content and an indication of fuel consumption of loosely compacted organic layer of 7 cm depth, weighing about 5 kg m^{-2}. DMC is also used as an indicator of the receptivity of the forest floor to ignition by lightning. (3) The Drought Code (DC) is a numerical rating of the average moisture content and an indication of seasonal drought effects on deep, compact, organic layers weighing about 25 kg m^{-2}. All the above fuel moisture codes are meant in the FWI System to model fuel moisture in a closed canopy mature pine stand.
- The three fire behaviour indices: (1) the Initial Spread Index (ISI) is a numerical rating of expected fire rate of spread. It combines the effect of wind and FFMC on rate of spread without the influence of variable quantities of fuel. (2) the Buildup Index (BUI) is a numerical rating of the total amount of fuel available for combustion that combines the DMC and the DC. (3) the Fire Weather Index (FWI) is a numerical rating of fire intensity that combines the ISI and the BUI.

The current version of the US National Fire Danger Rating System (NFDRS) was implemented in 1978 (Deeming *et al*., 1977), with optional revisions in 1988 (Burgan, 1988). NFDRS is a climatology-based system, and, as such, analysis of historical fire danger is required for proper interpretation and application of indices. The basis of both fire danger and fire behaviour systems is a physically based mathematical fire spread model (Rothermel, 1972). NFDRS provides a systematic way to integrate and interpret seasonal weather trends while fuel and terrain factors are essentially held constant. NFDRS uses daily weather observations and next day forecasts to produce fire danger indices. Weather readings are taken daily at the same time and location. Fuel moisture values are calculated for live grasses and shrub foliage and several size classes of dead fuels. The fuel moisture values are then used to calculate the indices. The NFDRS has three components: the Ignition Component (IC) which measures the flammability of live and dead fine forest fuels based on weather observations and fuel moisture measurements, and is expressed as the probability of a firebrand starting a sustainable fire; the fire Spread Component (SC), which is influenced most by the moisture content of fine dead fuel and the wind speed, and is calculated from the equations of the Rothermel fire spread model, and the Energy Release Component (ERC), which is a measure of the fire intensity and includes the contribution of heavy dead fuels in the combustion process. Thus, the SC reflects daily variations in fine fuel moisture and wind, while the ERC reflects longer term drying of the fuels.

The fire danger rating systems used in Australia are those developed by McArthur (1967) that predict ignition probability for forest and grassland fuels based on weather observations and fuel moisture content.

In southern European countries there is also a variety of methods in use for the evaluation of wildland fire danger:

- The French Method, known as Numerical Risk, proposed by Sol (1989) and Drouet and Sol (1990). It requires daily values of air temperature, relative humidity, cloud cover, wind velocity and an initial value of the water content of the soil. Additionally, Carrega (1991) has proposed an index from a statistical analysis of meteorological parameters associated with fire occurrence in the French Riviera.
- The Italian Method, IREPI, proposed by Bovio *et al.* (1984). This index estimates the loss of soil moisture due to actual evapotranspiration and compares it with the potential value of evapotranspiration, in order to compute the fire danger index. It requires the daily average values of air temperature, relative humidity, wind speed, insolation, and cumulative precipitation.
- The Portuguese Method is composed of a daily fire index and a cumulative index, the latter being a weighed sum of the daily indexes of the previous days, the weighting factor being a function of precipitation. For the calculation of the daily index it is necessary to measure the air temperature and relative humidity at 1200 LST. Wind speed and direction is taken into account in the final classification, according to local conditions (ICONA, 1988).

The Canadian FWI system has been adapted for use in a number of countries (Mexico, Indonesia, Malaysia, New Zealand) and these experiences have provided valuable insight in the adaptation of the system worldwide (Fogarty *et al.*, 1998). In these countries the FWI System is now operated nationally and is being used by fire and rescue agencies in order to develop and implement fire prevention, detection, and suppression plans (de Groot *et al.*, 2006).

The Southern Mediterranean countries of the European Union have requested that the Canadian FWI System be officially incorporated to the European Forest Risk Forecasting System (EFFRFS) in order to produce 1 and 3 day forest fire risk forecast maps at European and national level (Lopez *et al.*, 2002). Additionally, the FWI System is now being adapted to monitor drought and wildfire danger globally. The Global Fire Danger Rating System Project will be using daily weather data to produce fire weather maps from FWI system's codes and indices. When the system will be operational, the maps will be available online (GOFC – GOLD, 2003).

New Zealand adopted in 1978 the FWI System of the CFFDRS for use in exotic pine plantations with promising results (Pearce and Alexander, 1994). However, when in subsequent years the FWI System was used for fire-danger rating in a wide range of vegetation types in New Zealand, including shrublands, the system was not as successful. Thus, in 1992 a research program began to develop new models for indigenous fuel types, mainly shrublands, for use in the FWI (Fogarty *et al.*, 1998).

From 1981 to 1991, the Ontario Ministry of Natural Resources carried out a project to develop a model forest-fire management system, including a fire-danger rating system, in northeastern China (White and Rush, 1990). Many of the issues that the Canadian staff encountered in China were cultural. Decisions related to fire suppression had been reactive and were often made at a very senior level. Also, the use of Canadian-made electronics and weather instruments posed maintenance problems. The CFFDRS has also been used in broad-scale in Mexico, southeast Asia and Florida (USA), largely because of its simplicity and its strong interpretive products (Lee *et al.*, 2002). In Argentina, the National Fire Management Plan began to implement the FWI System in three pilot areas in 2001. In a review of the project, Taylor (2001) recommended that institutional mechanisms should be created and/or strengthened to enhance communication and cooperation between national and provincial agencies before implementing the system nationally. Fire Danger Rating Systems (FDRS) for Indonesia and Malaysia were developed by adapting components of the Canadian Forest Fire Danger Rating System, including the Canadian Forest Fire Weather Index (FWI) System and the Canadian Forest Fire Behaviour Prediction (FBP) System, to local vegetation, climate and fire regime conditions. These systems are now operated nationally and being used by fire and rescue agencies in order to develop and implement fire prevention, detection and suppression plans (de Groot *et al.*, 2006).

Several European countries, including Portugal and Spain, have also adopted parts of the CFFDRS. Viegas *et al.* (1999) found the FWI System components were well correlated with fire activity in southern Portugal, Spain, France and Italy although the vegetation and Mediterranean climate are much different than in Canada. Thus, it has been demonstrated that the CFFDRS has the potential for assessing fire danger in Southern Europe but, at the same time, it was recommended to conduct further tests of the system in more dry Mediterranean environments, such as Greece (Viegas *et al.*, 1994; Dimitrakopoulos and Bemmerzouk, 2003). The European Commission (2002) tested the FWI System in a pilot study during the 2000 and 2001 fire seasons in Spain, Portugal, France, Italy and Greece, with satisfactory results. However, the study identified the need for further evaluation of the FWI System in the Eastern Mediterranean countries (Greece) where, due to extended drought conditions, the duration of the fire season may be further prolonged compared to the West Mediterranean Basin. The European Forest Fire Information System (EFFIS) network has adopted the Fire Weather Index (FWI) as the method to assess the fire danger level in a harmonized way throughout Europe. The FWI algorithms have been slightly changed to better suit the remarkable differences in day length in European Union when going from the Mediterranean to the Boreal countries (Lopez *et al.*, 2002; San-Miguel-Ayanz *et al.*, 2003). Day-length adjustments were recently included in the calculation of the FWI for Australian conditions as well (Dowdy *et al.*, 2009).

The Mediterranean fire climate has the following distinctive characteristics (McCutchan, 1977):

- almost completely dry summers with concentration of the rainfall during the winter;
- extended periods of sunny weather and drought during the summer months, followed by periods of strong, desiccating winds, and,
- moderate marine-air influence throughout the year.

These conditions, in association with the flammable vegetation and the existing socio-economic conditions, result in many days of great fire potential and favourable conditions for large forest fires (Dimitrakopoulos and Panov, 2001).

The FWI system of the CFFDRS was designed for Canadian fuel and weather conditions. In other countries where environmental conditions may be quite different, all components of the CFFDRS may be neither necessary nor appropriate. However, because fuel wetting and drying processes are the same all over the world, and because fires respond to variation in the same physical influences of fuel, weather and topography, elements of the existing fire-danger systems can often be used.

The overall objective of this study was to investigate the basis and underlying relationships of the Canadian FWI System, in order to evaluate its potential use in a dry Mediterranean fire environment of the Eastern Mediterranean Basin (Greece). In order to achieve this, according to the selection criteria set by Fogarty *et al.* (1998), the following objectives were set:

- determine to what extent the FWI System components are correlated with measures of fire activity, such as fire occurrence and area burned;
- examine whether the fluctuations of the FWI System values can adequately reflect fire risk, as judged by actual fire occurrence, and,
- assess whether FWI System outputs can be used to predict relative fire environment components, such as fine fuel moisture and duff moisture.

Initially, the correlation between the indices of the FWI system and the number of wildfires and area burned was examined statistically for two fire seasons in Crete. Subsequently, logistic regression was applied to the FWI values and fire data to determine the range of the new fire danger classes, more appropriate for dry Mediterranean climates, based on the probability analysis of fire occurrence. Finally, the moisture content of the fine dead fuels and litter and duff layers was monitored with field samples and then correlated with the corresponding predictions of the FWI system, in order to test the validity of the system's outputs as predictors of fire activity in Mediterranean environments.

2. Methodology

2.1. Site description

The field tests were carried out at the Akrotiri region (approximately 7000 ha covered with sparse coniferous forest and evergreen shrublands), Northwestern Crete, Greece. The climate of the area is typically Mediterranean, characterized by a long dry summer period with a monotone wind regime, and precipitation chiefly occurring from September to May. During the last 20 years at the Akrotiti Airport (Chania, Crete, Greece) meteorological station the average annual precipitation was 690.4 mm. The rainy period, ranging from October to March, had an average precipitation of 615.6 mm, while the xerothermic period, extending from April to September, averaged only 74.7 mm. The average precipitation *per* month is as follows (in millimetres of rain): January 123, February 109, March 72, April 32, May 14, June 6.5, July 0.5, August 2.5, September 18, October 82, November 71, December 91. The first year of the study period had annual precipitation 719 mm (normal), while during the second year conditions of extreme drought were observed with total annual precipitation 225 mm. January and February are the coldest months, with an average temperature of 11.6 °C, and July and August the hottest months with average temperatures of 26.5 and 26.1 °C, respectively (Zaffran, 1990). The average temperature *per* month is as follows (in °C): January 11.6, February 11.6, March 13.2, April 16.3, May 20.1, June 24.5, July 26.5, August 26.1, September 23.3, October 19.4, November 16.1, December 13.1.

The sample site (35°31′N, 24°03′E) is covered by a typical Mediterranean forest of *Pinus halepensis* Mill. (Aleppo pine) with continuous litter (L) and duff (F) layer in the understory. The average depth of the forest floor layer (L and F layers mostly) was 5 cm. The L (litter) layer was on the surface of the forest floor and it was composed primarily of dry pine needles and twigs up to 0.5 cm in diameter, while the F (fermentation) layer was immediately under and it was composed primarily of partly decomposed litter and organic substance. The H (humus) layer, consisting solely of fully decomposed organic materials, was practically non-existent. Measurements were taken under a stand of approximately 50% crown closure, located at the bottom of a hillside with southwest aspect, at an altitude of 185 m above the sea level.

2.2. Field study methodology

The experiment was carried out during two fire seasons and measurements were conducted on a daily basis, at 1400 LST. Under the forest canopy, litter, duff and soil samples were collected. Every day, three sets of litter, duff and soil samples were each sampled at three separate locations, approximately 100 m from each other. The samples were placed into aluminium containers hermetically sealed, in order to prevent water loss through evaporation. The daily moisture value was determined as the average of the three values. The moisture content (MC) on an oven-dry weight basis (% o.d.w.), was calculated as follows (Viegas *et al*., 1992):

$$\text{MC (\% o.d.w.)} = [(\text{FW} - \text{DW})/\text{DW}] \times 100 \quad (1)$$

(FW: fresh weight, DW: dry weight after drying the sample for 24 h in an oven at 105 °C).

The litter, duff and mineral soil samples had approximately the following dimensions (L × W × D): 10 × 10 × 2–3 cm. Since the moisture content was the only parameter of interest, the samples were taken all together down to the mineral soil and then carefully separated from each other, according to the FWI system specifications. The weight of the litter and duff samples varied from 21 to 37 g, and that of soil 36 to 52 g. The moisture content of the upper mineral soil samples was correlated with the Drought Code (DC) values. Since the correlation was very poor, soil moisture was excluded as a factor from further analysis.

2.3. Data acquisition

The daily weather data (air temperature, air relative humidity, wind speed at 10 m, 24 h accumulated precipitation) necessary for computing the indices of the FWI System were obtained from the meteorological station of Akrotiti Airport, located in the same area at approximately 1.5 km from the experimental site.

Fire activity in the area was documented by the official wildland fire statistic reports of the Ministry of Agriculture, filled immediately after each fire by the Forest Service employees. These daily reports are very rigorous since, among other data, the location and area burned of every wildfire that occurred that day, is reported as well. Over 95% of total fires occur in the area during the period from April to October, thus resulting in a prolonged fire season, due to the favourable burning conditions (extended drought, high temperatures and low relative humidity) even during the autumn months (September, October). Almost all fires are deliberately set by shepherds who traditionally burn the area for 'range improvement', a practice very common in other areas of the Mediterranean Basin. The FWI values were calculated by the 'FWI Calculator' of the Canadian Forestry Service (Van Wagner and Pickett, 1985).

2.4. Statistical analysis

First, the degree of correlation of fire activity measures (daily number of fires and area burned) with the indices of the FWI system were statistically determined and tested for significance, to test the system's response in dry, Mediterranean environments.

Subsequently, the classification system of Ontario (Canada) was initially applied to classify FWI values to fire danger classes. However, this resulted in approximately 95% of the total number of days of each fire season to be classified in the 'Extreme' fire danger class. Thus, logistic regression was applied in order to define a reasonable range for each new fire danger class that would be appropriate for dry Mediterranean climates.

The logistic regression approach estimates the probability of occurrence of an event from a combination of fire weather and fire environmental factors. The logistic regression model has the following form (Hosmer and Lemeshow, 2000):

$$P(y_i = 1) = \frac{e^{g(x)}}{1 + e^{g(x)}} \quad (2)$$

being the logit given by the equation:

$$g(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_i x_i \quad (3)$$

where, $P(y_i = 1)$ is the probability of fire occurrence, x_i is the independent variable (FWI values), and β_ι are the coefficients estimated through the maximum likelihood method which maximizes the probability density as a function of the original set of data. A decision criterion of 0.5 was set as the threshold of fire occurrence. This cutoff value has been commonly used to discriminate the continuous model response in the interval between 0.0 and 1.0 (Wilson and Ferguson, 1986; Ryan and Reinhardt, 1988; Cruz *et al*., 2004)

The model was analysed through the rationality of the explanatory variables, the significance of the regression coefficients, and the discrimination capacity as measured by the area under a relative operating characteristic (ROC) curve (Wilson, 1987).

Finally, the measured moisture content values of the fine dead fuels and duff were correlated with those predicted by the FWI system, in order to test the system's validity in Mediterranean environments.

The statistical analysis was conducted by using the SPSS 12.0 statistical package (Norusis, 1997).

3. Results

3.1. Correlation of FWI system components with fire occurrence and area burned

Table I presents the correlation matrix between the components of the FWI System and the total number of fires and area burned during the two fire seasons in Crete, Greece. Fire occurrence was highly correlated with the DMC ($r = 0.89$), DC ($r = 0.78$), BUI ($r = 0.90$), and the FWI ($r = 0.59$). The correlations were statistically significant ($p < 0.05$). On the contrary, there was only moderate correlation between the components of the FWI System and the area burned: DMC ($r = 0.54$), BUI ($r = 0.52$), FWI ($r = 0.44$). None of these correlations was statistically significant. Stepwise multiple linear regression was also conducted but showed that the components of the FWI System explain a negligible fraction of the variance in the burned area.

Table I. Correlation matrix of the FWI system components and the number of fires and area burned for two fire seasons in the dry eastern Mediterranean environment of Crete, Greece.

	FFMC	DMC	DC	ISI	BUI	FWI
Area burned	0.42*	0.54	0.34	0.34	0.52	0.44
	(0.263)**	(0.1368)	(0.3748)	(0.3713)	(0.1472)	(0.2370)
Number of fires	0.50	0.89	0.78	0.35	0.90	0.59
	(0.1742)	(0.0014)	(0.0132)	(0.3578)	(0.0009)	(0.0316)
FFMC	1	0.73	0.24	0.94	0.70	0.94
	–	(0.0264)	(0.5289)	(0.0001)	(0.0365)	(0.0002)
DMC	–	1	0.74	0.63	0.99	0.81
	–	–	(0.0228)	(0.0698)	(0.0001)	(0.0070)
DC	–	–	1	0.16	0.78	0.49
	–	–	–	(0.6761)	(0.0123)	(0.1757)
ISI	–	–	–	1	0.60	0.93
	–	–	–	–	(0.0883)	(0.0002)
BUI	–	–	–	–	1	0.81
	–	–	–	–	–	(0.0082)
FWI	–	–	–	–	–	1

* Correlation coefficient, ** p values in parentheses, level of significance $p < 0.05$.

3.2. FWI index and fire risk

The FWI index (FWI) is a nonlinear, basically empirical combination of the ISI and BUI (Stocks *et al*., 1987). To facilitate the use of the FWI, the classification system (0–5 Low, 6–12 Moderate, 13–24 High, >24 Extreme) established in Ontario, Canada, was initially adopted in our study too (Alexander *et al*., 1996). The classification of the FWI values of this study in the Canadian fire danger classes resulted in about 95% of all days during the summer months to be classified in the 'Extreme' fire danger class. Such a high number of 'Extreme' fire danger days diminishes the operational use of a fire danger system. Thus, logistic regression was applied to the FWI values in order to define new fire danger classes more appropriate for the dry Mediterranean climate of Crete. The estimated parameters, the standard error and the significance level of the logistic model are presented in Table II. The logistic model yielded an area under the ROC curve of 0.86. Overall, the model correctly predicted fire occurrence and non-fire occurrence incidents in 79.17% of all cases (Table III). Table IV presents the new fire danger classes derived from the probability analysis of fire occurrence and FWI values with the logistic model. The probability of fire occurrence in Table IV corresponds to the adjacent FWI value. The upper bound level (95%) of the FWI value was used as the criterion to define the upper end value of each new fire danger class based on the logistic model, as follows (probability of fire occurrence): 0–38 (0.01–0.29) Low, 39–48 (0.30–0.59) Moderate, 49–60 (0.60–0.89) High, >60 (0.90–0.99) Extreme.

The distribution of the number of fires to the new fire danger classes showed that 74% of total fires occurred in days with 'High' and 'Extreme' FWI values during the second year, as opposed to 28% during the first year (Table V). The first year only the 5.5% of the number of days was classified into 'High' and 'Extreme' fire danger class, while the second year 32.4% of the number of days were classified into 'High' and 'Extreme' fire danger class. This is expected, since the second year studied experienced prolonged drought conditions.

The comparison of the two fire seasons studied shows that the second year had a prolonged fire season (extending through October) as compared to the first year, where the fire season was restricted from May to August (Table VI). This is due to the difference in the amount of total and autumn precipitation (719 and 120.6 mm for the first year against 225 and 25.0 mm for the second year, respectively), which was perfectly reflected in the FWI values.

Table II. Estimated parameters and statistics associated with the probabilistic fire occurrence logistic model[a].

Variable	Value	Standard error	Wald chi-square	Pr > Chi^2
β_0	−6.405	0.824	60.482	<0.0001
β_1	0.141	0.018	59.846	<0.0001

[a] Equation form: Probability of fire occurrence $= 1/[1 + \exp(-6.405 + 0.141 \times \mathrm{FWI})]$

Table III. Classification table comparing observed and predicted fire occurrence probability through the application of the logistic model to the data.

Observed	Predicted		
	Non−fire occurrence	Fire occurrence	Percent correct (%)
Non−fire occurrence	104	23	81.89
Fire occurrence	27	86	76.11
	Overall		79.17

Table IV. Fire danger classes (based on FWI values) derived from probability analysis with logistic regression of fire occurrence data during two fire seasons (one normal and one dry) in the eastern Mediterranean environment of Crete, Greece.

Probability of fire occurrence	FWI value	Lower bound 95%	Upper bound 95%	Fire danger class
0.01	12.796	1.839	19.456	Low
0.05	24.467	17.289	28.950	
0.10	29.750	24.202	33.328	
0.20	35.484	31.552	38.232	
0.30	39.295	36.245	41.684	Moderate
0.40	42.419	39.873	44.733	
0.50	45.286	42.959	47.774	
0.60	48.153	45.805	51.054	High
0.70	51.277	48.694	54.843	
0.80	55.088	52.030	59.651	
0.90	60.822	56.854	67.081	Extreme
0.95	66.105	61.200	74.027	
0.99	77.776	70.665	89.505	

Table V. Distribution of fire occurrence and number of days in the FWI danger classes during the 2 years studied.

FWI classes	0–38 low	39–48 moderate	49–59 high	>60 extreme
% days (first year)	81.5	13	4	1.5
% fire occurrence (first year)	42	30	22	6
% days (second year)	54.9	12.7	20.7	11.7
% fire occurrence (secondyear)	16.3	9.4	40.6	33.7

3.3. Correlation between measured litter and duff moisture contents and FWI predictions

Two Fuel Moisture Codes of the FWI System were studied: the Fine Fuel Moisture Code (FFMC) and the Duff Moisture Code (DMC). Because the codes are inversely related to fuel moisture, fine fuel moisture (FFM) is obtained by subtracting FFMC from 101%. The following exponential transformation of Van Wagner's (1987) original equation is given by Stocks *et al.* (1987) to convert DMC to duff moisture (DM):

$$DM = e^{-\frac{(DMC - 244.72)}{43.43}} + 20 \tag{4}$$

The descriptive statistics listed in Table VII indicate that the observed and the predicted values of FFM are very similar. Daily variations of observed and predicted fine fuel moisture content values are presented in Figure 1. Correlation between observed and predicted values gave an $r = 0.89$ at the 95% confidence level. A t-test of paired observed and predicted litter moisture content values showed that they were not significantly different at 95% confidence level.

Table VI. Distribution of number of days and number of fires per month to the new FWI fire danger classes, during the 2 years studied.

FWI values danger class	0–38 low	39–48 moderate	49–60 high	>60 extreme
First year				
March	31[a](1)[b]	0 (0)	0 (0)	0 (0)
April	30 (0)	0 (0)	0 (0)	0 (0)
May	28 (2)	2 (0)	0 (0)	1 (0)
June	16 (1)	8 (2)	4 (1)	2 (1)
July	17 (2)	10 (0)	4 (1)	0 (0)
August	17 (4)	10 (1)	3 (3)	1 (1)
September	24 (13)	4 (3)	2 (2)	0 (0)
October	31 (6)	0 (0)	0 (0)	0 (0)
November	29 (0)	1 (0)	0 (0)	0 (0)
Second year				
March	31 (0)	0 (0)	0 (0)	0 (0)
April	30 (0)	0 (0)	0 (0)	0 (0)
May	21 (1)	5 (0)	2 (1)	3 (1)
June	5 (0)	8 (1)	7 (4)	10 (3)
July	2 (1)	5 (1)	13 (5)	11 (2)
August	3 (0)	6 (3)	18 (12)	4 (3)
September	7 (5)	6 (2)	13 (16)	4 (5)
October	27 (13)	2 (1)	2 (0)	0 (0)
November	25 (0)	3 (0)	2 (0)	0 (0)

[a] Number of days.
[b] Number of fires.

All predicted duff moisture descriptive statistical parameters were higher than observed (Table VII). Daily variations of observed and predicted duff moisture content values are presented in Figure 2. A t-test of paired observed and predicted litter moisture content values showed that they were significantly different at 95% confidence level.

4. Discussion

Harrington *et al.* (1983) also found poor correlation between area burned and components of the FWI System for nine Canadian provinces (explained variance between 12 and 33%), and concluded that this does not necessarily reflect the accuracy and usefulness of the fire indexes. It could be anticipated that the weather, as expressed by the components of the System, would explain a significant fraction of variance in burned area. However, these components will never perfectly express the 'Fire–Weather' relationship, because a fraction of the explained variance would be lost by aggregation of burned area by province and due to limitations on the number and locations of weather stations. Area burned, the most evident characteristic of fire statistics, is affected by many factors including: wind speed, number of simultaneous fires, variations in fire accessibility, fire control organizational efficiency, the nature and extent of fuel, topography and weather. On the other hand, the high correlation between fire occurrence and FWI values in Crete, where most

Table VII. Descriptive statistics for observed and predicted fine fuel moisture (FFM) and duff moisture (DM) values.

	Observed FFM	Predicted FFM	Observed DM	Predicted DM
Mean	11.9	11.6	21.37	37.15
Maximum	67.2	66.2	144.6	275
Minimum	6.2	4.5	11.46	20
Standard deviation	7.24	6.28	21.32	49.12
Coefficient of variation (%)	0.61	0.54	0.99	1.32
Range	60.9	61.7	133.14	255.36

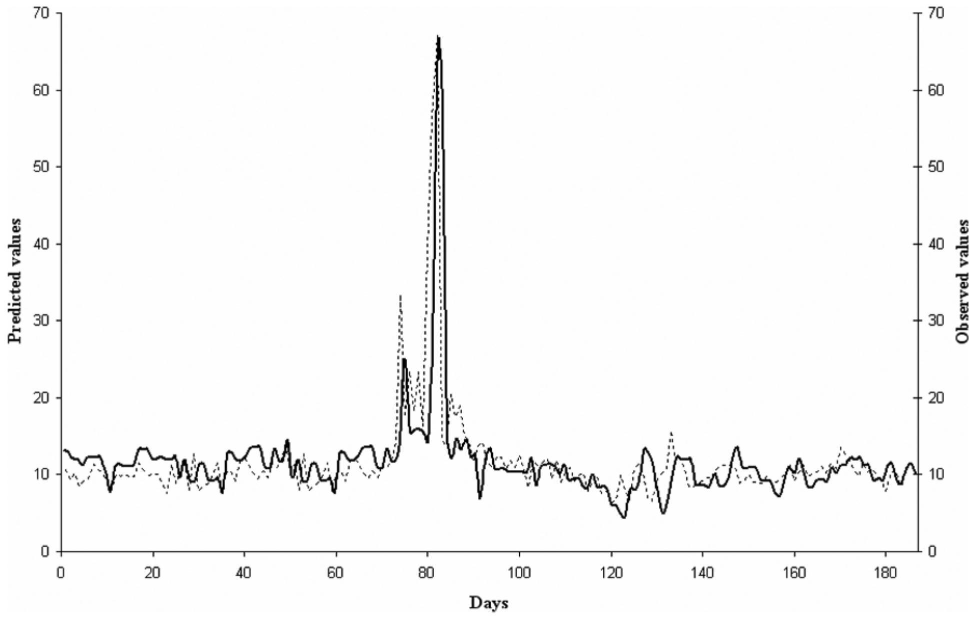

Figure 1. Comparison of observed (dots) and predicted (line) fine fuel (litter) moisture values (FFM) derived from the FFMC. [Observed FFM $= -0.058 + 1.028\times$ (Predicted FFM), $r = 0.89$].

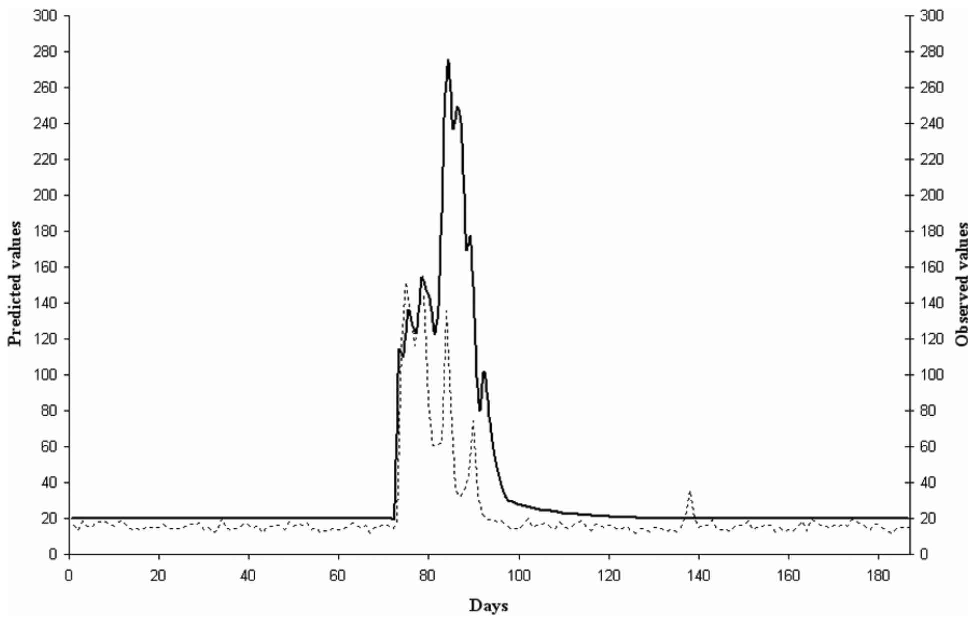

Figure 2. Comparison of observed (dots) and predicted (line) duff moisture content values (DM) derived from the DMC. [Observed DM $= 9.45 + 0.32\times$ (Predicted DM), $r = 0.75$].

fires are deliberately set by shepherds for pasture management, could be attributed to the arsonists' strategy to start fires during days of high fire risk (i.e. periods of prolonged drought, high temperatures and low relative humidity that dry out the fine fuels, thus increasing their ignitability).

The new classification of fire danger classes successfully reflected wildfire risk, since actual fire occurrence was substantially higher during days of 'High' and 'Extreme' fire risk. For both fire seasons, May represents the threshold where FWI values reach the 'High' and 'Extreme' classes. For the two fire seasons studied, September is the month with the most fires. This is due to the fact that no rain has occurred since late spring, creating a prolonged drought period (at least 4 months), coinciding with the period of the highest temperature and solar radiation. This results in a pronounced water stress level for the vegetation, since the soil water storage capacity is nearly exhausted. This situation is exacerbated when strong south winds accompanied by very low relative humidity greatly increase the fire danger in the Mediterranean region (McCutchan, 1977).

The potential range of the FFMC is 0–101. In the present case, FFMC values range from 10.3 to 96.5, with a mean value for the two fire seasons of 84.95. It is worth noting that 90% of the observed values lay above 75, while the value of 65 is considered as the threshold at which the probability of fire occurrence begins to increase exponentially (Wotton, 2009). It should be emphasized that the relationship between the FFMC and flammability is highly non-linear such that small variations in FFMC at the high end of the scale represent important variations in flammability (Harrington *et al.*, 1983). The results suggest that a great majority of days during the fire season demonstrate conditions suitable for fire occurrence and a non-significant increase in the FFMC values can reflect big differences in the flammability of the fuels making them burn readily and with ferocity. Regarding the observed *versus* predicted fine fuel moisture content (FFM) values, the general behaviour of the two sets of points is similar throughout the seasons, although there seems to be a better agreement during the summer period, when there is no rainfall. This indicates the sensitivity of the model to daily changes of air relative humidity, temperature and wind speed. Immediately after rain, the model seems to slightly underpredict the moisture content, but resumes very quickly. Given the practical importance of the lower range of moisture content as regard to fire risk assessment, the slight underestimation in the higher range does not present a major shortcoming.

The duff layer in the Mediterranean ecosystems is significantly less in terms of depth and quantity than that of the original Canadian environment. In the lower range, the predictions are limited when the observed duff moisture content drops below 20%. This is due to the fact that the DMC was set to an equilibrium moisture content of 20%, meaning that the lowest value the model can predict should be at least equal to 20. However, during the summer period the duff moisture in Mediterranean pine forests drops well below this threshold. The integration of a 'minimum equilibrium moisture content' specific for the Mediterranean conditions, which could be daily calculated from the maximum temperature and minimum relative humidity, would help overcome this weakness.

The Drought Code (DC) was initially included in the study, but it was later deleted because the correlations with observed duff and soil moisture contents were not significant at the 95% confidence level. The DC represents deep layers of compact organic matter that rarely exist in the Mediterranean forests. In our study, samples were taken from the top soil layer which is subjected to surface weather conditions. Therefore, the DC could not be properly evaluated by this study.

5. Conclusions

The Fire Weather Index (FWI), tested at the dry Mediterranean-type environment of Crete (Greece), demonstrated several aptitudes for its potential use as a meteorological fire danger index in regions of mediterranean climate. The Duff Moisture Code (DMC), the Drought Code (DC), the Buildup Index (BUI) and the FWI were highly correlated with fire occurrence but only moderately with area burned. The new FWI classes successfully reflected actual fire risk during the two fire seasons studied, one with normal precipitation and the other under extreme drought conditions.

High correlation was found between the observed and predicted fine fuel (litter) moisture ($r = 0.89$). This indicates the sensitivity of the model, not only to the sharp fluctuations of moisture content after rain, but also to daily variations of relative humidity, temperature and wind speed during the summer dry period.

A delayed response was observed between the observed and predicted values of duff moisture. Further modification in the high range of the predicted duff moisture content values would decrease the reaction time to rain occurrence, and the integration of a 'minimum equilibrium moisture content', appropriate to Mediterranean-type ecosystems, would improve predictions in the low range. The DC was poorly correlated to litter, duff, and soil moisture contents.

Although a variety of meteorological conditions occurred during the study period, it is unlikely that the full range of fuel moisture and weather variations that are observed in a typical dry Mediterranean environment were covered. The new FWI fire danger classes have a potential for fire risk assessment in eastern (dry) Mediterranean environments. However, for future research, a similar study should be carried out over a broad scale and for a longer time period, in order to be more conclusive regarding the adaptability of the FWI to Mediterranean-type environments.

List of symbols, quantities and units used in equations and text

Symbol or abbreviation	Definition and units
BUI	Buildup Index
CFFDRS	Canadian Forest Fire Danger Rating
DC	Drought code
DMC	Duff moisture code
DW	Dry weight (kg)
EFFRFS	European Forest Fire Risk Forecasting System
FFMC	Fine fuel moisture code
FWI	Fire weather index
FW	Fresh weight (kg)
ISI	Initial spread index
MC	Moisture content (% ovendry weight basis)

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